

Department of Economics, University of Vermont

June 2026 · Live Online

`thinkingwithagents.github.io`

This Hour



1. A Few Ideas

Teaching prep is **production** — agents make it cheap. Your judgement stays the scarce input.

~10 min, slides

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Papers in → a ready-to-teach lecture out: notes, slides, discussion prompts, exam questions. Skill + live demo.

~25 min, mostly live

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One prompt → an interactive tool your students play with **inside your LMS**. Live demo + embed walkthrough.

~20 min, live

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Most of this hour happens **live in the terminal**. Every prompt we run is on a slide so you can copy as we go.

Teaching Prep Is Production — and Production Got Cheap



What agents now produce in minutes

- ▶ Lecture notes & slide decks from papers
- ▶ Problem sets, exam questions, rubrics
- ▶ Interactive tools, simulations, dashboards
- ▶ Reading lists — found *and* downloaded

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What stayed yours

- ▶ Is it **correct**? (numbers, citations, claims)
- ▶ Is it **right for your students**? (level, pacing)
- ▶ Is it **what you'd stand behind** in class?

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Review everything **before** the classroom does.

Today's Two Demos = Two Ends of a Spectrum



A skill: lecture-builder

A reusable **playbook**: parse the paper in chunks → structured extraction → notes + deck, every time.

- ▶ Same file layout, same quality, repeatable
- ▶ Has an **audience knob**: undergrad vs. grad
- ▶ Worth building for workflows you repeat all semester

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One good prompt: the dashboard

No skill, no setup — a single well-specified prompt produces a polished interactive tool.

- ▶ Right for **one-off artifacts**
- ▶ The craft is in the **specification**
- ▶ If you start repeating it → promote it to a skill

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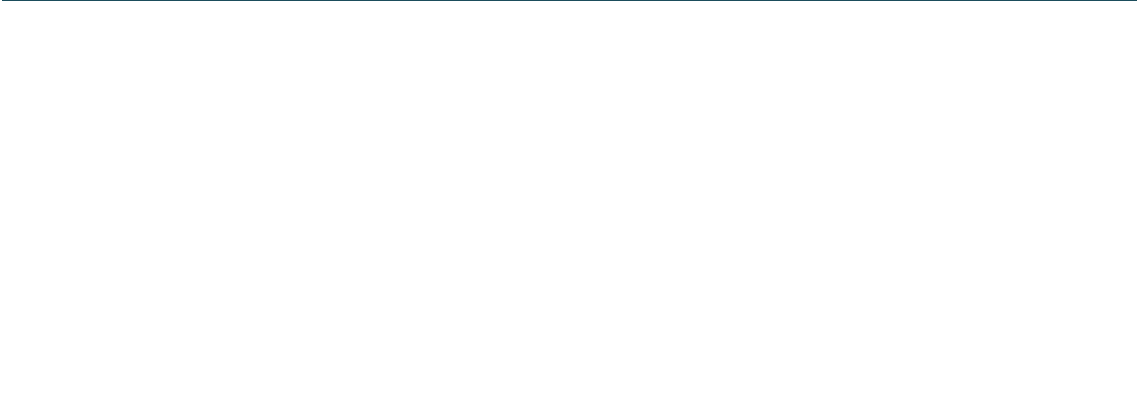


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The decision rule: **Repeat it? Skill it. One-off? Prompt it.**



lecture-builder — What It Is



Document in → lecture out

- ▶ Takes a paper (PDF/DOCX/URL) — parses it **in chunks**, so a 100-page paper doesn't blow up the context window
- ▶ Extracts claims, findings, methods, discussion prompts, exam questions into one structured file
- ▶ Writes `lecture_notes.md` + a Beamer deck from that **same** extraction — notes and slides can't drift apart

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The audience knob

Undergraduate: intuition first, jargon defined, open-ended discussion.

Graduate: identification at a craft level, critique & replication-style questions.

Flip it with one word in the prompt.

Slide titles are **takeaways**, not topics: “Productivity up 15%” beats “Results.”

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Today's run: an undergrad lecture on “**AI and the Labor Market**” — a topic your students are already asking about.

Step 1: One Prompt Finds (and Fetches) the Reading List

```
> claude
```

```
I'm preparing an undergraduate lecture on how AI is affecting the economy, especially the labor market.
```

1. Search broadly for the most influential papers (2023-2026): employment, wages, tasks, productivity, inequality.
2. Build an annotated reading list (authors, venue, one-line takeaway) and save it as papers/reading_list.md.
3. Download the open-access PDFs (arXiv, NBER, journal sites) into papers/ and verify each file is a real PDF.

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In our test run this took **~3 minutes**: 6 papers found,
6 PDFs downloaded & verified, reading list written.

What Came Back: Six Papers, Annotated, On Disk

Paper	One-line takeaway (as annotated by the agent)
Brynjolfsson, Li & Raymond (<i>QJE</i> 2025)	AI assistant for 5,172 support agents: productivity +15% — +30%+ for novices, ≈ 0 for the most experienced
Eloundou et al. (2023, <i>Science</i> 2024)	$\sim 80\%$ of workers have $\geq 10\%$ of tasks LLM-exposed; high-wage jobs most exposed
Acemoglu (<i>Economic Policy</i> 2024)	Task model + Hulten: 10-year TFP gain $\leq \mathbf{0.66\%}$ — far below the hype
Autor (NBER 2024)	The optimist case: AI can extend expertise and rebuild middle-class jobs
Humlum & Vestergaard (2025)	Danish admin data, DiD: precise null on earnings & hours two years post-adoption
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A built-in classroom debate: micro gains are large (row 1), exposure is broad (row 2), yet aggregate effects are ≈ 0 so far (row 5).

Step 2: Feed the Papers to lecture-builder

Now build me a 50-minute undergraduate lecture from the papers in papers/.

Use "Generative AI at Work" (Brynjolfsson, Li & Raymond) as the anchor paper and weave the others in as context: exposure, the macro debate, and the early aggregate evidence.

I want lecture notes, a Beamer slide deck, in-class discussion prompts, and candidate exam questions.

Audience: principles/intermediate undergraduates - intuition first, define jargon, takeaway-style slide titles.

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▶ What the skill does under the hood

parse_document.py chunks each PDF into sections (38 for the anchor) → reads selectively, not wholesale → extraction.json (claims, findings + *where in the paper*, methods, questions) → notes + deck generated from that single source of truth.

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Output of our test run: **15-frame compiled Beamer deck + 11-section lecture notes** + 5 discussion prompts + 6 exam questions (short answer, MC, essay) — a TA could run the section from the notes alone.

Live: From a Paper to a Lecture, Start to Finish

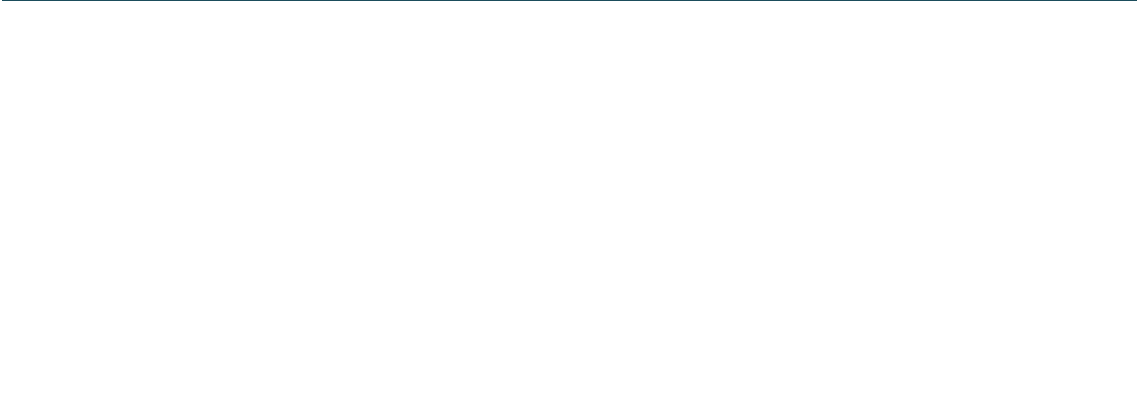
Pick a paper. Watch it become
notes + slides + discussion prompts + exam questions.

parse · extract · notes · deck · compile

Watch for: the chunked parse (no 100-page context dump) · claims-not-topics in `extraction.json` · the audience knob doing real work.

Then the 60-second edit: “make slide 8 graduate-level” — regenerate one piece, not the whole deck.

Teaching a paper next week? Drop its title or arXiv link in the chat — if time allows, we'll run yours.



The Idea: Interactive Tools as Single Files



Why one self-contained HTML file

- ▶ Everything inside: graph, sliders, quiz — **no internet, no installs, no accounts**
- ▶ Opens in any browser, on laptops *and* phones
- ▶ Embeds in **every** LMS (Canvas, Blackboard, Brightspace, Moodle) — next slides
- ▶ Collects no student data; can't break your course page

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Why it teaches

Students don't watch the comparative statics — they **cause** them. Predict first (quiz mode), then watch the graph act it out.

Think “a PDF that happens to be interactive” — that's the mental model for your LMS, too.

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Worked example: a **supply & demand explorer** for principles of micro — but the recipe is course-agnostic.

The Prompt (the Craft Is in the Specification)

```
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```

```
Build an interactive supply-and-demand explorer for my principles of microeconomics class:
```

- one self-contained HTML file, no external libraries, must work offline and inside an LMS iframe
- sliders to shift demand and supply, plus preset scenario buttons (incomes rise, input costs rise, per-unit tax on sellers, better technology); show the original curves dashed after a shift
- live readout of equilibrium price and quantity, with up/down arrows vs. the starting point
- a "quiz me" mode: ask a shift question, let the student answer, then animate the correct shift
- clean modern look, readable on phones, axis labels, no walls of text

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Every line buys robustness: **self-contained** (no library = nothing to break in an iframe) · **offline** · **phone-readable** · **screenshot** so you verify by *seeing*. This is the whole spec — there is no step 2.

What It Built (Our Test Run, Unedited)

dashboard-shifted-crop.png

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dashboard-shifted-crop.png



Out of the box

- ▶ 6 preset scenarios + free sliders
- ▶ Quiz mode: predict, then watch
- ▶ Equilibrium deltas with arrows
- ▶ Deep links: ?s=-22 opens pre-shifted — link it from a homework question

One file, ~400 lines, zero dependencies. Built and verified in minutes.

Putting It in Your LMS (No Coding, Two Paths)



Path A — host once, embed anywhere (recommended)

Tell the agent: *"Publish this to GitHub Pages and give me the link."* Then paste **one line** into any LMS editor's embed/HTML button:

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<iframe src="https://you.github.io/econ-dashboards/supply-demand-explorerer.html" width="100%" height="700"></iframe>
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LMS	Where the embed button lives
Brightspace (D2L)	Content → Create a File → Insert Stuff → Enter Embed Code
Canvas	Pages → Edit → Insert → Embed (or the <> HTML editor)
Blackboard	Build Content → Item → <> source-code button (Ultra: Document → Add HTML)

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Path B — no hosting at all: upload the .html straight into the LMS (Brightspace renders it interactively as a topic; Canvas/Blackboard via Files), or just email it — it works offline by double-clicking.

Why the iframe trick is universal: LMS editors *strip* pasted JavaScript (security), but they all *allow* iframes.

Not Just Intro Micro: The Recipe Is Course-Agnostic

Same prompt skeleton — swap the concept, the controls, and the quiz:

Micro & policy

Tax incidence & DWL · price ceilings/floors · elasticity sandbox · monopoly markup · game-theory payoff matrices

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Health: insurance & moral hazard · Labor: minimum-wage DiD explorer · Trade: comparative-advantage PPFs · Enviro: cap-and-trade

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The constraint is no longer “can I build it?”
— it’s “**what do I want students to *do*?**”

Pedagogy first: predict → manipulate → explain. The tool just makes that loop cheap.

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Dry-run everything. Our test run caught a real bug — before your 9am section did.

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Department of Economics, University of Vermont

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